

Coloring random subgraphs

OPG-827

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

If G is a graph and $p \in [0, 1]$, we let G_p denote a subgraph of G where each edge of G appears in G_p with independently with probability p .

Problem Does there exist a constant c so that $\mathbb{E}(\chi(G_{1/2})) > c \frac{\chi(G)}{\log \chi(G)}$?